

2. A herbal energy drink

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Software

Starts in 28 min

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Argument: A specific brand's herbal energy drink was given to a group of 50 consumers and a study was conducted to prove the benefits of the drink on the mood of a person. Consumers were put in time bound situations that and then given a glass of the drink to help them focus. The test results were out and 80 percent of the consumers agreed the energy drink in fact did help with their productivity. Hence, it can be proven that the specific brand's energy drink on which the survey was conducted is the best energy drink in the market.

The above given argument's reasoning is most vulnerable to criticism on the grounds that it fails to consider which of the given choices?

Answer Options

Select any one option

[Clear Answer](#)

- ☐ Any of the consumers that participated in the survey are allergic to some herbs in the drink.
- ☐ The herbs in the energy drink is not very harmful to the health of the consumer.
- ☐ The price of the energy drink when compared to other brands is high.
- ☐ Customers had the opportunity to taste the energy drink from various other brands as well. ✓

5. The Christmas carols

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Software

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You are organizing a program for Christmas and want to form a choir group. You want the choir group to consist of 3 boys and 2 girls from a group of 7 boys and 10 girls.

In how many ways can you form the choir group if one boy and one girl refuse to sing together?

Answer Options

Select any one option

[Clear Answer](#)☐ 1460 ways☐ 1540 ways☒ 1440 ways☐ 1600 ways[Submit test](#)

4. Trains A and B

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Software

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Two trains A and B are running on parallel tracks in the same direction. The length of A is 65 m and it is running at a speed of 34km/hr. The length of B is 70 m and it is running at a speed of 25 km/hr.

How long will it take for both the trains to pass each other completely from the moment they met?

Answer Options

Select any one option

[Clear Answer](#)☒ 54 seconds☐ 56 seconds☐ 50 seconds☐ None of these

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1. Value of sum ✓

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The elements of an array **A** of size **n** are assigned to random values distributed uniformly in the range -2 to 2. After doing so, the following loop is executed.

```
for (sum=0, i=0; i if (A[i] > 0)
```

```
sum += A[i];
```

What is the expected value of **sum** at the end of the loop if **n** is large?

Answer Options

Select any one option

Clear Answer

☐ 4*n☐ 2*n☐ n☒ 0

Answer Saved



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2. Managing deadlocks ✓

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Read the following statements about deadlocks carefully and choose the correct option.

A: Circular wait is a necessary condition for deadlocks to occur.

B: Circular wait is a sufficient condition for deadlocks to occur.

Answer Options

Select any one option

Clear Answer

☐ A and B are both true.☒ A is true, B is false.☐ A is false, B is true.☐ A and B are both false.

Answer Saved



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5. A semaphore variable

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You are performing the following operations on a certain semaphore variable.

1. Signal
2. Control
3. Release
4. Wait

Which of these operations are permitted in the given scenario?

Answer Options

Select any one option

Clear Answer

☐ Only signal and control☐ Only signal and release☒ Only wait and signal☐ None of these

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4. The LCALL instruction

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You are using the **LCALL** instruction to call a subroutine in an 8051 microprocessor. What can be said about the **RET** instruction in this scenario?

1. The RET instruction transfers the control back to the caller after finishing the execution of the subroutine.
2. Every subroutine uses RET as the last instruction.

Answer Options

Select any one option

Clear Answer

☐ Only 1☐ Only 2☒ Both 1 and 2☐ Neither 1 nor 2

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6. The increment operator

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What is the output of the following C program?

```
#include <stdio.h>
void solve() {
    int a = 3;
    int res = ++a + a++ + a++ + ++a;
    printf("%d", res);
}
int main() {
    solve();
    return 0;
}
```

Answer Options

Select any one option

[Clear Answer](#)☐ 12☐ 24☒ 21☐ 18

Think 20



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3. Two stages of a pipeline

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1 2 3
4 5 6
7 8 9
10

A certain register is used to hold the intermediate output between two stages in a pipeline.

What are these registers called?

Answer Options

Select any one option

[Clear Answer](#)

☐ Batch

☒ Buffer or Latch

☐ Buffer or Batch

☐ Latch or Batch

Answer Saved

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7. Processes accessing the same data ✓

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Suppose two processes A and B access the same data concurrently and the outcome of the execution depends on the order in which the access of resource takes place.

Which of the following choices accurately depicts such a scenario?

Answer Options

Select any one option

[Clear Answer](#)☐ Critical section☒ Race condition☐ Circular wait☐ All the above☐ None of these

Answer Saved



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8. Processing an algebraic series

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Three numbers (2, 3, 5) in an algebraic series are processed inside a queue **q** and stack **s** which are empty initially.

Identify the correct output after the following operations executes.

```
q.enqueue(3);  
q.enqueue(5);  
q.enqueue(2);  
s.push(q.dequeue());  
s.push(q.dequeue());  
s.pop();
```

Answer Options

Select any one option

Clear Answer

☐ q.front() returns 3 and the stack is empty☐ q.front() returns 2 and the stack is empty☐ q.front() returns 3 and s.top() returns 5☒ q.front() returns 2 and s.top() returns 3

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9. Sorting an integer data ✓

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An integer data set needs to be sorted in ascending order. It is given that the user is free to augment the data set by adding additional elements when the sorting operation is already taking place and the sorting process needs to account for this data.

Which of the following sorting algorithms would be appropriate for usage in the given scenario?

Answer Options

Select any one option

[Clear Answer](#)☐ Selection sort☐ Merge sort☐ Heap Sort☒ Insertion Sort

Answer Saved



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10. A weird fruit

You have a weird fruit with you that you can either sell as it is or you can divide it magically into 3 parts $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and sell them. It is given that the division performed is an integer division and initially the weight of the fruit is n units.

You have to determine the maximum weight of the fruit that you can obtain from the magical division procedure. You are given the following pseudocode for performing this operation.

Solve(n):

if($n==0$):

return 0

return X

Analyse the code and choose the correct option that fills the blank X.

Note:

Example: $n=2$, if you divide it you will get 1,0,0 weights i.e 1 unit, so max is 2 unit of weight

Example: $n=12$, if you divide it you will get 6,4,3 so total is 13 (you don't divide them further).

Answer Options

Select any one option

Clear Answer

☒ X: $\max(n, \text{solve}(n/2) + \text{solve}(n/3) + \text{solve}(n/4))$

☐ X: $\text{solve}(n/2) + \text{solve}(n/3) + \text{solve}(n/4)$

☐ X: n

☐ X: $\max(n+3, \text{solve}(n/2) * \text{solve}(n/3) * \text{solve}(n/4))$



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